

Cryptographic Algorithms

Cryptography

CRYPTOGRAPHIC ALGORITHMS

Cryptographic algorithms can be categorised into three classes: ***Hash functions***, ***Symmetric*** and ***Asymmetric*** key algorithms.

Hash functions

A hash function is a cryptographic algorithm which is used to transform large random size data to small fixed size data. Hashing condenses a message into an irreversible fixed-length value, digest, or *hash*.

Symmetric key algorithm

Symmetric-key algorithms also referred as secret-key algorithms use a single cryptographic key for encryption and decryption purposes.

Asymmetric key algorithm

Sometimes referred as public key cryptography. They use two mathematically associated keys known as public and private keys. Private key for encryption and Public key for decryption.

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Cryptographic algorithms also differ in the **amount of data** that is processed at a time. Some algorithms use a stream cipher that takes one character and replaces it with one character while other algorithms make use of a block cipher. In 2007, another type was introduced, sponge function.

A **Stream cipher** works on one character at a time.

A **block cipher** manipulates an entire block of plaintext at one time. The plaintext message is divided into separate blocks of 8 to 16 bytes, and then each block is encrypted independently. For additional security, the blocks can be randomised.

A **sponge function** takes as input a string of any length and returns a string of any requested variable length. This function repeatedly applies a process on the input that has been padded with additional characters until all characters are used (absorbed in the sponge).